

Student Gradebook And Result Analyzer

Project Proposal



Session:2025-2029

Course: CSC-102 — Programming Fundamentals

Contributors

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Submitted To

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1. Project Overview

The Student Gradebook and Result Analyzer is a menu-driven console application developed as a Complex Computing Problem (CCP). The system is designed to manage student academic records and perform performance analysis for instructors.

The application allows a teacher to:

- View student records in a structured tabular format
 - Search, update, delete, and sort student data
 - Analyze academic performance using statistical reports
 - Export processed results and analytics to a CSV file for persistence and future use
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2. Problem Understanding and Assumptions

Due to partially ill-defined requirements, the following grading policies and system constraints were explicitly defined to ensure correctness and consistency.

2.1 Grading Scale and Mapping

A standard 4.0 GPA scale, commonly used in Pakistani engineering universities, is used to convert percentage scores into Grade Points.

Percentage Range	Letter Grade	Grade Points
90% and above	A+	4.0
86% – 89%	A	3.9
82% – 85%	A-	3.7

78% – 81%	B+	3.3
74% – 77%	B	3.0
70% – 73%	B-	2.7
66% – 69%	C+	2.3
62% – 65%	C	2.0
58% – 61%	C-	1.7
54% – 57%	D+	1.3
50% – 53%	D	1.0
Below 50%	F	0.0

2.2 GPA Calculation Formula

The Semester Grade Point Average (SGPA) is calculated using weighted credit hours:

$$SGPA = \frac{\sum_{i=1}^n (\text{GradePoints}_i \times \text{CreditHours}_i)}{\sum_{i=1}^n \text{CreditHours}_i}$$

2.3 System and CSV Assumptions (A8)

To ensure modularity and interdependence, the following assumptions were established:

- **Course Load:**
Each student must be enrolled in exactly (user-defined) number of subjects, with a total semester load between 15–18 credit hours.

- **Component Weightage:**

Each course is evaluated out of 100 marks, distributed as:

- Quizzes (2): 20%
- Assignments: 10%
- Mid-Term Exam: 30%
- Final Exam: 40%

ID	Name	Course Name	Cr.Hrs	Quiz1	Quiz2	Assignment	Mids	Finals	..
560	Ammar Bin Yasir	PF	3	10	8	10	28	35	..

Sample CSV Structure

CSV File Specifications

- The first row contains column headers
- `ID` serves as the primary key (integer)
- `Name` contains the student's full name
- `Course Name` contains the name of the course the student is taking
- `Cr.Hrs` are the number of credit hours in that course
- The subsequent columns are for the marks of that student in that course for his quizzes, assignments, mids, and finals.
- The file is loaded at program startup and updated upon any modification
- **Pass/Fail Criteria:**
 - A student is considered passed if:
 - $SGPA \geq 1.0$
 - No F grade in any core subject

3. Program Design

The application follows a modular design, where each component contributes to overall system accuracy and reliability.

3.1 Main Program Flow

The `main()` function provides a menu-driven user interface and coordinates all program operations.

Execution Flow:

1. Initialize an array of `Student` structures
2. Load student data from `students.csv` using `loadRecords()`.
3. Enter a `do-while` loop with menu-based options
4. Perform selected operations via switch cases

Available Operations:

1. Display all student records
 2. Search for a student by ID
 3. Update student marks
 4. Sort records (ascending or descending order)
 5. View analytics and statistics
 - Topper list
 - Subject-wise averages
 - Pass/Fail ratio
 6. Export reports to a user-specified file path
 7. Save changes and exit
-

3.2 Data Structures

- `struct Course`
Stores assessment components, total marks, and assigned grade.
 - `struct Student`
Stores Student ID, Name, an array of `Course` structures, and computed SGPA.
-

3.3 Core Functions

- **File Operations**

`-saveDataToFile()` — Serializes the current array of student objects into a structured CSV file for long-term storage.

`-loadDataFromFile()` — Populates the program's data structures by reading student records and grades from an existing CSV file.

`-saveReportToFile()` — Exports a formatted academic performance report to an external text file for printing or distribution.

- **Display and Reporting**

`-displayStudents()` — Iterates through the student list to provide a summary view of all enrolled students.

`-displayStudentProfile()` — Displays a detailed view of a specific student's personal information and individual course scores.

`-reportGeneration()` — Orchestrates the creation of comprehensive academic summaries, including averages and top performers.

- **Data Management**

`-sortStudents()` — Reorders the student records based on specific criteria in either ascending or descending order.

- **Academic Logic & Calculations**

`-calculateStudentResults()` — Triggers the comprehensive calculation logic for total marks, grade points, and SGPA for a specific student.

`-courseTotalMarks()` — Sums up weighted assessments (assignments, exams, etc.) to determine the final score for a single course.

`-getGradePoint()` — Maps a numerical course score to a standardized grade point value (e.g., 3.7).

`-getGradeLetter()` — Assigns the appropriate alphabetical grade (e.g., "B+") based on total marks earned in a course.

`-calculateSGPA()` — Computes the Semester Grade Point Average by weighting grade points against course credit hours.

`-getSemesterGrade()` — Determines the final academic standing or descriptive grade based on the calculated SGPA.

- **Utilities & Analytics**

`-formatFloat()` — A helper utility that rounds and converts floating-point numbers into strings with specific decimal precision.

`-printrow()` — A UI helper that generates consistent horizontal dividers and borders for tabular terminal output.

`-printCell()` — Handles the alignment and padding of data within a specific column width for table consistency.

`-topper()` — Identifies the highest-scoring student in a specific subject and returns both their score and index.

`-subjectAverage()` — Calculates the mean score across all students for a specific course or assessment.

`-passFailRatio()` — Computes the number of students who met the passing criteria versus those who did not for a specific subject.

4. How to Run the Program

1. Place `data.csv` in the same directory as the executable
 2. Compile the source code using any standard C++ compiler
 3. Run the program and interact using the menu-driven interface
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5. Testing and Edge Cases

- Empty file handling
 - Invalid student ID input
 - Boundary GPA values (e.g., 1.0, 4.0)
 - Duplicate IDs prevention
 - Sorting with identical GPAs
-

6. Sample Generated Reports

(Included in exported CSV files)

```
=====
||                               Student Management System                               ||
=====

--- Main Menu ---
1. Display the table of all (current) records.
2. Search a Student based on their ID.
3. Update Student Marks.
4. Sort the Student list in (ACS/DESC).
5. View Analytics & Statistics.
6. Export Reports in a file.
7. Save and Exit.
Select Option: |
```


ID	Name	PF	DM	Cal-1	AICT	AP	AP-Lab	PF-Lab	AICT-Lab	SGPA	Grade
588	Muhammad Khuzaib	72.00	71.00	68.00	72.00	79.00	75.00	86.00	66.00	2.77	B-
560	Ammar Bin Yasir	77.00	72.00	93.00	84.00	63.00	69.00	83.00	87.00	3.15	B
574	Maheen Noor	83.00	68.00	80.00	77.00	65.00	76.00	82.00	78.00	2.99	B-
554	Abdullah Maqsood	96.00	80.00	82.00	78.00	84.00	76.00	78.00	71.00	3.50	B+
558	Ali Hassan	84.00	71.00	82.00	89.00	70.00	79.00	91.00	78.00	3.38	B+
566	Fatima Ajmal	75.00	95.00	73.00	83.00	80.00	89.00	75.00	79.00	3.33	B+
583	Muhammad Asad Mukhtar	71.00	85.00	91.00	74.00	72.00	89.00	76.00	75.00	3.28	B
569	Hareem Fatima	85.00	68.00	65.00	83.00	79.00	68.00	63.00	92.00	2.89	B-
590	Muhammad Ramzan	84.00	81.00	67.00	90.00	62.00	78.00	80.00	86.00	3.15	B
578	Muhammad Abdul Quyyum Javed	86.00	80.00	89.00	74.00	72.00	75.00	71.00	71.00	3.32	B+
581	Muhammad Ali	76.00	81.00	84.00	81.00	80.00	87.00	77.00	71.00	3.30	B+
580	Muhammad Ahmed Kashif	75.00	77.00	87.00	76.00	67.00	66.00	84.00	77.00	3.08	B
605	Zainab Abdullah	74.00	78.00	78.00	82.00	70.00	81.00	84.00	85.00	3.27	B
602	Syed Muhammad Abbas Ali	65.00	83.00	72.00	82.00	67.00	96.00	70.00	80.00	2.95	B-
553	Abdullah Abbas	75.00	90.00	79.00	90.00	74.00	88.00	91.00	90.00	3.55	B+
551	Abdul Hanan	73.00	72.00	85.00	90.00	83.00	89.00	78.00	93.00	3.37	B+
556	Ahmed Bilal Qazi	92.00	88.00	78.00	75.00	93.00	71.00	93.00	77.00	3.58	B+
573	M Bilal Ahmad	90.00	92.00	88.00	89.00	91.00	83.00	83.00	85.00	3.91	A
577	Momina Iftikhar	84.00	80.00	76.00	71.00	72.00	69.00	77.00	83.00	3.11	B
601	Syed Ahmad Mujtaba	77.00	85.00	75.00	80.00	77.00	81.00	92.00	78.00	3.27	B
568	Haider Ali	71.00	82.00	80.00	79.00	80.00	96.00	85.00	82.00	3.16	B+
579	Muhammad Ahmad	66.00	84.00	84.00	79.00	68.00	88.00	90.00	72.00	3.18	B
599	Roshaan Shahzad	72.00	78.00	82.00	69.00	89.00	67.00	79.00	71.00	3.11	B
575	Maryam Amjad	70.00	92.00	81.00	84.00	85.00	77.00	73.00	91.00	3.41	B+
562	Ashna Sami Butt	71.00	91.00	71.00	67.00	78.00	88.00	82.00	84.00	3.17	B
565	Eiman	85.00	71.00	93.00	79.00	72.00	72.00	66.00	69.00	3.16	B
597	Neha rehan	67.00	77.00	71.00	72.00	74.00	71.00	82.00	80.00	2.82	B-
591	Muhammad Rayan Zubair	74.00	72.00	78.00	76.00	81.00	82.00	67.00	74.00	3.04	B
600	Shahzaib Riaz	87.00	82.00	75.00	85.00	77.00	86.00	72.00	75.00	3.43	B+
563	Ayesha	84.00	72.00	81.00	77.00	73.00	88.00	71.00	77.00	3.13	B
604	Usman Ali	80.00	75.00	74.00	76.00	77.00	72.00	81.00	85.00	3.10	B
555	AbdulRehman	81.00	82.00	89.00	81.00	68.00	85.00	89.00	79.00	3.42	B+
592	Muhammad Siddique Raza	77.00	76.00	90.00	75.00	82.00	79.00	72.00	90.00	3.34	B+
557	Ali Abbas	72.00	83.00	82.00	86.00	70.00	80.00	70.00	76.00	3.28	B
567	Hafiz Muhammad Abdullah Aamir	81.00	76.00	70.00	74.00	76.00	76.00	89.00	78.00	3.08	B
594	Muhammad Usman	79.00	77.00	92.00	72.00	67.00	87.00	91.00	85.00	3.28	B
589	Muhammad Muzamil	69.00	82.00	81.00	76.00	88.00	83.00	91.00	81.00	3.29	B
552	Abdul Sabokh	74.00	81.00	84.00	80.00	73.00	78.00	82.00	69.00	3.21	B
559	Ali Talha	79.00	68.00	83.00	64.00	84.00	68.00	93.00	76.00	3.04	B
593	Muhammad Toseef	87.00	79.00	85.00	75.00	83.00	69.00	93.00	75.00	3.46	B+
585	Muhammad Azhar Riaz	81.00	89.00	81.00	71.00	78.00	86.00	78.00	79.00	3.38	B+
603	Talha Sajid	76.00	77.00	87.00	73.00	81.00	90.00	72.00	73.00	3.19	B
596	Muneeb ur Rahman	85.00	81.00	73.00	68.00	71.00	79.00	76.00	91.00	3.09	B
595	Muhammad Zaid	67.00	74.00	69.00	78.00	92.00	76.00	78.00	82.00	2.96	B-
564	Bilal Ahmed	90.00	84.00	65.00	83.00	82.00	62.00	83.00	83.00	3.33	B+
570	Hifza Kiran	78.00	72.00	84.00	83.00	76.00	70.00	83.00	73.00	3.23	B
571	Javeria Nadeem	81.00	77.00	71.00	74.00	75.00	84.00	89.00	78.00	3.12	B
582	Muhammad Anas Majeed	83.00	65.00	79.00	83.00	71.00	64.00	79.00	77.00	3.01	B
587	Muhammad Hassan	83.00	78.00	72.00	89.00	88.00	78.00	88.00	86.00	3.49	B+
598	Rehan Ahmed	97.00	85.00	77.00	85.00	77.00	80.00	77.00	75.00	3.42	B+
584	Muhammad Awais Shehzad	64.00	87.00	68.00	93.00	79.00	65.00	81.00	86.00	3.02	B
586	Muhammad Gulraiz	79.00	80.00	68.00	68.00	80.00	77.00	71.00	79.00	2.93	B-
561	Arham Shakeel	84.00	74.00	72.00	76.00	83.00	70.00	83.00	86.00	3.24	B
576	Mehar Waseem	77.00	65.00	79.00	87.00	72.00	86.00	83.00	84.00	3.09	B
572	Khalil Ur Rehman	73.00	87.00	81.00	78.00	82.00	80.00	73.00	83.00	3.34	B+

--- Main Menu ---

1. Display the table of all (current) records.
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3. Update Student Marks.
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5. View Analytics & Statistics.
6. Export Reports in a file.
7. Save and Exit.

Select Option: |

Enter the id of the Student you want to search: 562

Name: Ashna Sami Butt
ID: 562

Course	Quiz 1	Quiz 2	Assign.	Mids	Finals	Total	CGPA	Grade
PF	8.00	6.00	9.00	22.00	26.00	71.00	2.70	B-
DM	10.00	5.00	10.00	26.00	40.00	91.00	4.00	A+
Cal-1	10.00	10.00	9.00	14.00	28.00	71.00	2.70	B-
AICT	6.00	7.00	8.00	14.00	32.00	67.00	2.30	C+
AP	8.00	9.00	10.00	25.00	26.00	78.00	3.30	B+
AP-Lab	7.00	5.00	9.00	30.00	37.00	88.00	3.90	A
PF-Lab	10.00	5.00	10.00	24.00	33.00	82.00	3.70	A-
AICT-Lab	8.00	10.00	9.00	29.00	28.00	84.00	3.70	A-
							3.17	B

--- Main Menu ---

1. Display the table of all (current) records.
 2. Search a Student based on their ID.
 3. Update Student Marks.
 4. Sort the Student list in (ACS/DESC).
 5. View Analytics & Statistics.
 6. Export Reports in a file.
 7. Save and Exit.
- Select Option: |

Enter the id of the Student you want to update: 562

Select which Subject you want to update.

1. PF
2. DM
3. Cal-1
4. AICT
5. AP
6. AP-Lab
7. PF-Lab
8. AICT-Lab

1

Select which marks you want to update.

1. Quiz 1
2. Quiz 2
3. Assignment
4. Mids
5. Finals

5

Enter Updated marks (0-40): 40

Data saved successfully to: data.csv

Name: Ashna Sami Butt
ID: 562

Course	Quiz 1	Quiz 2	Assign.	Mids	Finals	Total	CGPA	Grade
PF	8.00	6.00	9.00	22.00	40.00	85.00	3.70	A-
DM	10.00	5.00	10.00	26.00	40.00	91.00	4.00	A+
Cal-1	10.00	10.00	9.00	14.00	28.00	71.00	2.70	B-
AICT	6.00	7.00	8.00	14.00	32.00	67.00	2.30	C+
AP	8.00	9.00	10.00	25.00	26.00	78.00	3.30	B+
AP-Lab	7.00	5.00	9.00	30.00	37.00	88.00	3.90	A
PF-Lab	10.00	5.00	10.00	24.00	33.00	82.00	3.70	A-
AICT-Lab	8.00	10.00	9.00	29.00	28.00	84.00	3.70	A-
							3.36	B+

--- Main Menu ---

1. Display the table of all (current) records.
 2. Search a Student based on their ID.
 3. Update Student Marks.
 4. Sort the Student list in (ACS/DESC).
 5. View Analytics & Statistics.
 6. Export Reports in a file.
 7. Save and Exit.
- Select Option: |

```

1. Sort in ascending order
2. Sort in descending order
1
Enter the file name in to store the sorted data to.
data.csv
1. Sort based on ID
2. Sort Based on Semester GPA
1

```

ID	Name	PF	DM	Cal-1	AICT	AP	AP-Lab	PF-Lab	AICT-Lab	SGPA	Grade
551	Abdul Hanan	73.00	72.00	85.00	90.00	83.00	89.00	78.00	93.00	3.37	B+
552	Abdul Sabokh	74.00	81.00	84.00	80.00	73.00	78.00	82.00	69.00	3.21	B
553	Abdullah Abbas	75.00	90.00	79.00	90.00	74.00	88.00	91.00	90.00	3.55	B+
554	Abdullah Maqsood	96.00	80.00	82.00	78.00	84.00	76.00	78.00	71.00	3.50	B+
555	AbdulRehman	81.00	82.00	89.00	81.00	60.00	85.00	89.00	79.00	3.42	B+
556	Ahmed Bilal Qazi	92.00	88.00	78.00	75.00	93.00	71.00	93.00	77.00	3.58	B+
557	Ali Abbas	72.00	83.00	82.00	86.00	70.00	80.00	70.00	76.00	3.28	B
558	Ali Hassan	84.00	71.00	82.00	89.00	70.00	79.00	91.00	78.00	3.38	B+
559	Ali Talha	79.00	68.00	83.00	64.00	84.00	68.00	93.00	76.00	3.04	B
560	Ammar Bin Yasir	77.00	72.00	93.00	84.00	63.00	69.00	83.00	87.00	3.15	B
561	Arham Shakeel	84.00	74.00	72.00	76.00	83.00	70.00	83.00	86.00	3.24	B
562	Ashna Sami Butt	85.00	91.00	71.00	67.00	78.00	88.00	82.00	84.00	3.36	B+
563	Ayesha	84.00	72.00	81.00	77.00	73.00	88.00	71.00	77.00	3.13	B
564	Bilal Ahmed	90.00	84.00	65.00	83.00	82.00	62.00	83.00	83.00	3.33	B+
565	Eiman	85.00	71.00	93.00	79.00	72.00	72.00	66.00	69.00	3.16	B
566	Fatima Ajmal	75.00	95.00	73.00	83.00	80.00	89.00	75.00	79.00	3.33	B+
567	Hafiz Muhammad Abdullah Aamir	81.00	76.00	70.00	74.00	76.00	76.00	89.00	78.00	3.08	B
568	Haider Ali	71.00	82.00	80.00	79.00	80.00	96.00	85.00	82.00	3.36	B+
569	Hareem Fatima	85.00	68.00	65.00	83.00	79.00	68.00	63.00	92.00	2.89	B-
570	Hifza Kiran	78.00	72.00	84.00	83.00	76.00	70.00	83.00	73.00	3.23	B
571	Javeria Nadeem	81.00	77.00	71.00	74.00	75.00	84.00	89.00	78.00	3.12	B
572	Khalil Ur Rehman	73.00	87.00	81.00	78.00	82.00	80.00	73.00	83.00	3.34	B+
573	M Bilal Ahmad	90.00	92.00	88.00	89.00	91.00	83.00	83.00	85.00	3.91	A
574	Maheen Noor	83.00	68.00	80.00	77.00	65.00	76.00	82.00	78.00	2.99	B-
575	Maryam Amjad	70.00	92.00	81.00	84.00	85.00	77.00	73.00	91.00	3.41	B+
576	Mehar Waseem	77.00	65.00	79.00	87.00	72.00	86.00	83.00	84.00	3.09	B
577	Mosina Iftikhar	84.00	80.00	76.00	71.00	72.00	69.00	77.00	83.00	3.11	B
578	Muhammad Abdul Quyyum Javed	86.00	80.00	89.00	74.00	72.00	75.00	71.00	71.00	3.32	B+
579	Muhammad Ahmad	66.00	84.00	84.00	79.00	68.00	88.00	90.00	72.00	3.18	B
580	Muhammad Ahmed Mashif	75.00	77.00	87.00	76.00	67.00	66.00	84.00	77.00	3.08	B
581	Muhammad Ali	76.00	81.00	84.00	81.00	80.00	87.00	77.00	71.00	3.30	B+
582	Muhammad Anas Majeed	83.00	65.00	79.00	83.00	71.00	64.00	79.00	77.00	3.01	B
583	Muhammad Asad Mukhtar	71.00	85.00	91.00	74.00	72.00	89.00	76.00	75.00	3.28	B
584	Muhammad Awais Shehzad	64.00	87.00	68.00	93.00	79.00	65.00	81.00	86.00	3.02	B
585	Muhammad Azhar Riaz	81.00	89.00	81.00	71.00	78.00	86.00	78.00	79.00	3.38	B+
586	Muhammad Gulraiz	79.00	80.00	68.00	68.00	80.00	77.00	71.00	79.00	2.93	B-
587	Muhammad Hassan	83.00	78.00	72.00	89.00	88.00	78.00	88.00	86.00	3.49	B+
588	Muhammad Khuzaim	72.00	71.00	68.00	72.00	79.00	75.00	86.00	66.00	2.77	B-
589	Muhammad Muzaail	69.00	82.00	81.00	76.00	88.00	83.00	91.00	81.00	3.29	B
590	Muhammad Ramzan	84.00	81.00	67.00	90.00	62.00	78.00	80.00	86.00	3.15	B
591	Muhammad Rayan Zubair	74.00	72.00	78.00	76.00	81.00	82.00	67.00	74.00	3.04	B
592	Muhammad Siddique Raza	77.00	76.00	90.00	75.00	82.00	79.00	72.00	90.00	3.34	B+
593	Muhammad Toseef	87.00	79.00	85.00	75.00	83.00	69.00	93.00	75.00	3.46	B+
594	Muhammad Usman	79.00	77.00	92.00	72.00	67.00	87.00	91.00	85.00	3.28	B
595	Muhammad Zaid	67.00	74.00	69.00	78.00	92.00	76.00	78.00	82.00	2.96	B-
596	Muneeb ur Rahman	85.00	81.00	73.00	68.00	71.00	79.00	76.00	91.00	3.09	B
597	Neha rehan	67.00	77.00	71.00	72.00	74.00	71.00	82.00	80.00	2.82	B-
598	Rehan Ahmed	97.00	85.00	77.00	85.00	77.00	80.00	77.00	75.00	3.42	B+
599	Roshaan Shahzad	72.00	78.00	82.00	69.00	89.00	67.00	79.00	71.00	3.11	B
600	Shahzaib Riaz	87.00	82.00	75.00	85.00	77.00	86.00	72.00	75.00	3.43	B+
601	Syed Ahmad Mujtaba	77.00	85.00	75.00	80.00	77.00	81.00	92.00	78.00	3.27	B
602	Syed Muhammad Abbas Ali	65.00	83.00	72.00	82.00	67.00	96.00	70.00	80.00	2.95	B-
603	Talha Sajid	76.00	77.00	87.00	73.00	81.00	90.00	72.00	73.00	3.19	B
604	Usman Ali	80.00	75.00	74.00	76.00	77.00	72.00	81.00	85.00	3.10	B
605	Zainab Abdullah	74.00	78.00	78.00	82.00	70.00	81.00	84.00	85.00	3.27	B

Data saved successfully to: data.csv

```

--- Main Menu ---
1. Display the table of all (current) records.
2. Search a Student based on their ID.
3. Update Student Marks.
4. Sort the Student list in (ACS/DESC).
5. View Analytics & Statistics.
6. Export Reports in a file.
7. Save and Exit.
Select Option: |

```

SEMESTER ANALYTICS REPORT			
Subject	Topper(score)	Class Average	Pass/Fail Ratio
PF	Rehan Ahmed(97.00)	78.76	55/0
DM	Fatima Ajmal(95.00)	79.13	55/0
Cal-1	Ammar Bin Yasir(93.00)	78.98	55/0
AICT	Muhammad Awais Shehzad(93.00)	78.82	55/0
AP	Ahmed Bilal Qazi(93.00)	76.95	55/0
AP-Lab	Haider Ali(96.00)	78.62	55/0
PF-Lab	Ahmed Bilal Qazi(93.00)	80.47	55/0
AICT-Lab	Abdul Hanan(93.00)	79.75	55/0

```

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7. Save and Exit.
Select Option: |

```

```
--- Main Menu ---
1. Display the table of all (current) records.
2. Search a Student based on their ID.
3. Update Student Marks.
4. Sort the Student list in (ACS/DESC).
5. View Analytics & Statistics.
6. Export Reports in a file.
7. Save and Exit.
Select Option: 6
Enter the file name in to store the Semester Analytics and Reports to.
data.csv
Data saved successfully to: data.csv

--- Main Menu ---
1. Display the table of all (current) records.
2. Search a Student based on their ID.
3. Update Student Marks.
4. Sort the Student list in (ACS/DESC).
5. View Analytics & Statistics.
6. Export Reports in a file.
7. Save and Exit.
Select Option: |
```

```
--- Main Menu ---
1. Display the table of all (current) records.
2. Search a Student based on their ID.
3. Update Student Marks.
4. Sort the Student list in (ACS/DESC).
5. View Analytics & Statistics.
6. Export Reports in a file.
7. Save and Exit.
Select Option: 7
Saving and Exiting...
Data saved successfully to: data.csv

C:\Users\ashna\OneDrive\Documents\Desktop\PF Theory Project\PF-theory-project\x64\Debug\PF-theory-project.exe (process 16472) exited with code 0 (0x0).
Press any key to close this window . . .|
```
